Uni[MASK]: Unified Inference in Sequential Decision Problems

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NEURAL INFORMATION PROCESSING SYSTEMS

Problem

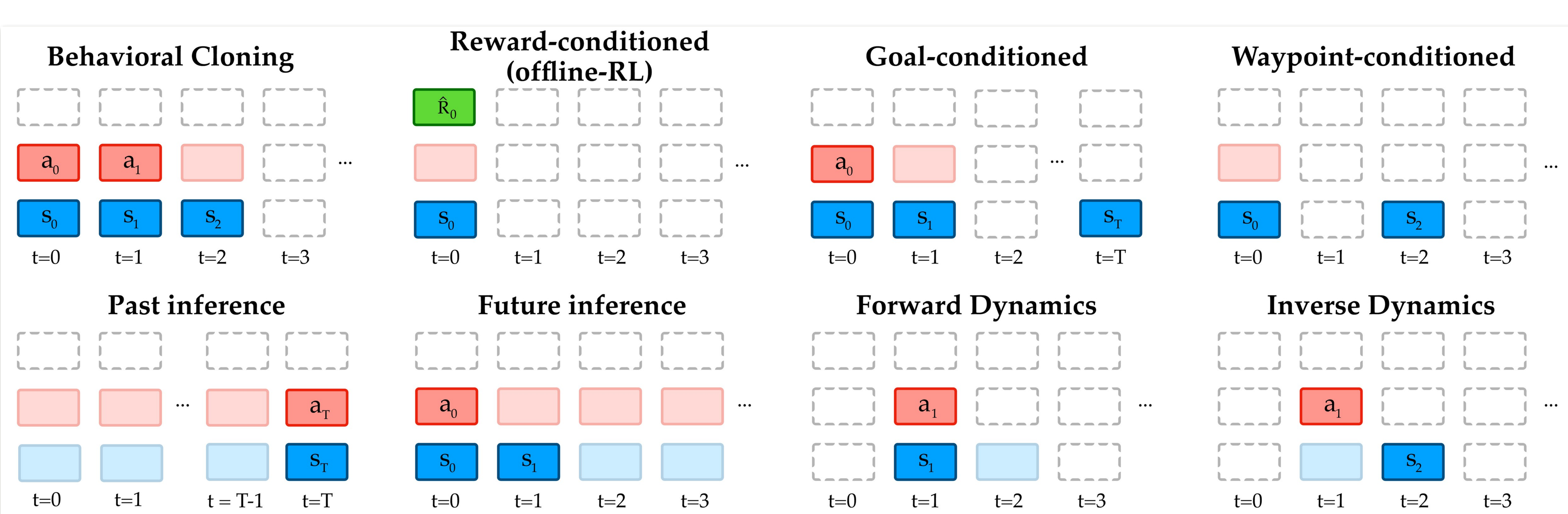
Sequential decision problems are usually solved **one task at a time** — a separate model for behavior cloning, offline RL, dynamics prediction, or goal conditioning — even though the tasks share deep structure.

> *All of these inferences run over the **same trajectory** of states, actions and returns — and could share representations — yet each is trained in isolation.*

Motivation

- Masked language modeling (BERT) trains models to predict randomly masked tokens, yielding bidirectional representations that transfer to many downstream tasks.
- Decision-making inferences are just different **maskings of a trajectory**: mask the last action given prior states and actions, and predicting it is exactly a behavior-cloning inference.

A single masked-prediction objective can express the whole family — behavior cloning, offline RL, dynamics, and goal or waypoint conditioning — with no task-specific heads.



Uni[MASK] framework: eight decision-making tasks — BC, offline RL, goal / waypoint conditioning, past / future inference, and forward / inverse dynamics — written as masking schemes over one trajectory (solid = input, translucent = predicted).

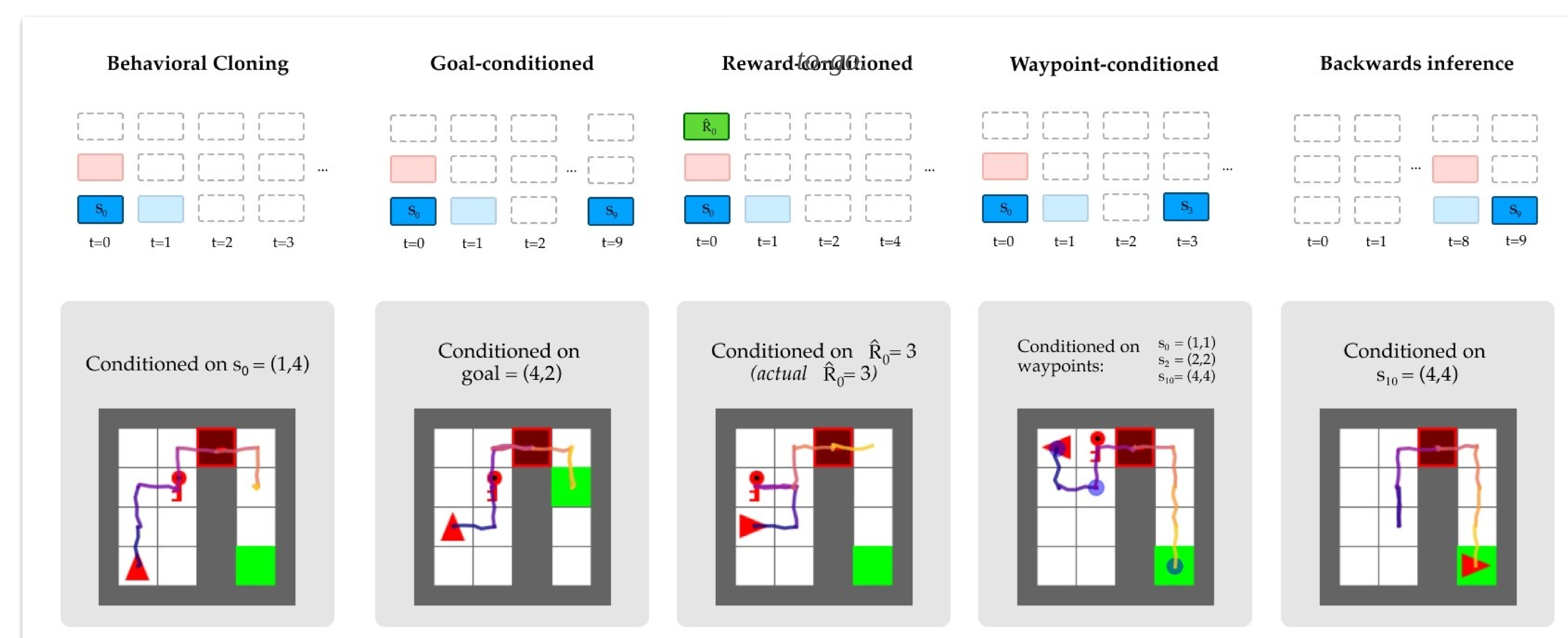
Method

- Tokenize a trajectory into per-timestep **state, action, and property** (return-to-go) tokens.
- A **masking scheme** picks inputs vs. predicted outputs; a bidirectional **BERT transformer** predicts the masked ones.

$$P(a_t \mid s_{0:t},\, a_{0:t-1})$$ havior cloning behavior cloning

$P(a_2 \mid s_{0:2,T},\, a_{0:1},\, \hat{R}_0)$

Every task is a masking scheme, e.g. *conditions an action on a goal state + return-*



One Uni[MASK] model executes five inference tasks in MiniGrid — behavioral cloning, goal-, reward-, waypoint-conditioned, and backwards inference — each just a different masking. Four training regimes (single-task, multi-task, random-mask, +fine-tune) are compared.

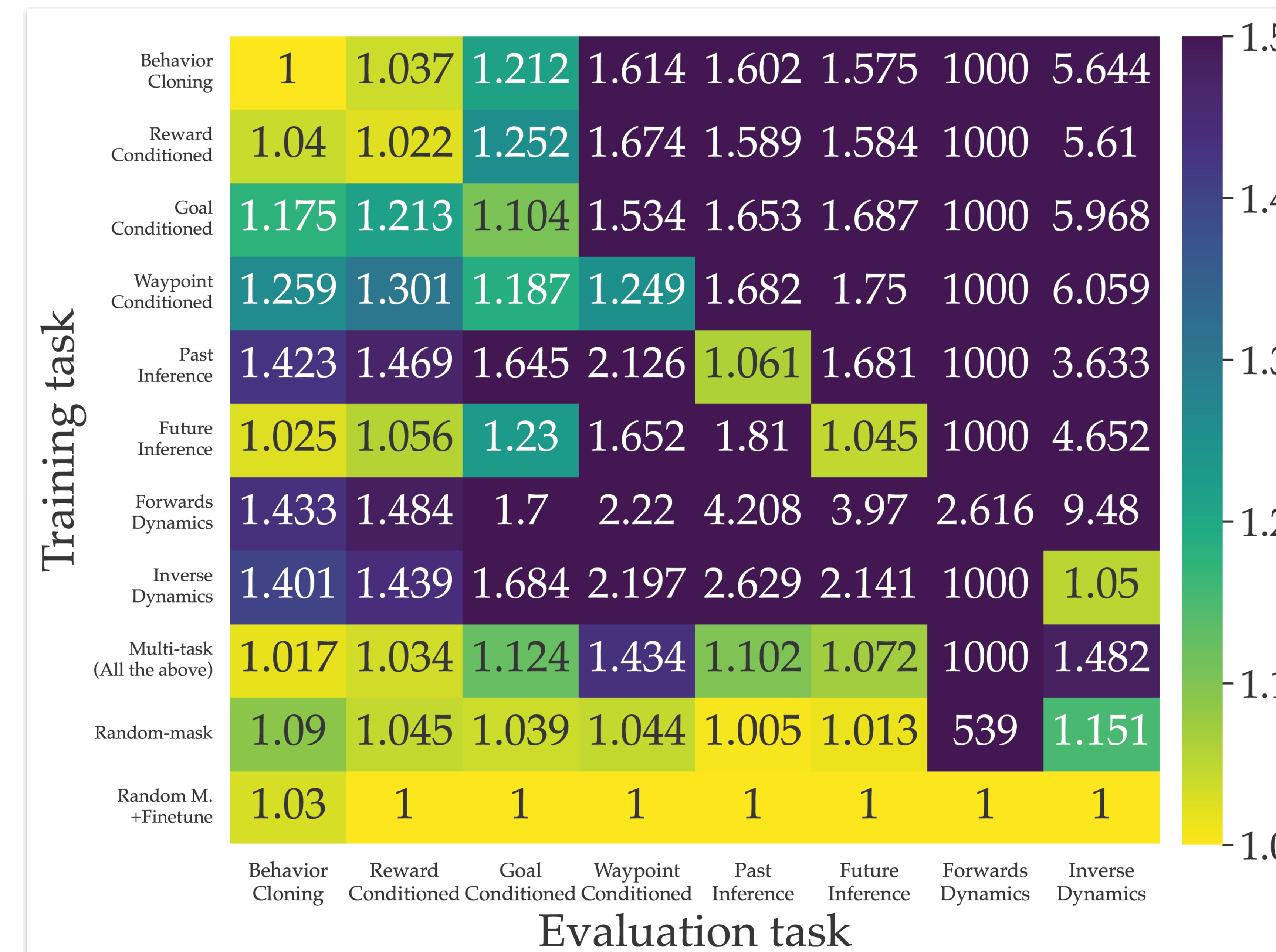
Takeaway

Many sequential-decision tasks are just different maskings of one trajectory — so one masked-prediction model replaces a zoo of specialized ones, and random-mask pretraining + fine-tuning generally beats task-specific training.

BERT-style backbones shine at short contexts but struggle to generate over longer sequences — GPT-style backbones with random masking are a promising next step.

Key Results

MAZE2D REWARD · CL 5	BC	RC
Decision Transformer	1.13	1.49
Uni[MASK] + finetune	**2.73**	**2.73**



MiniGrid cross-task validation loss (relative to best). Random-mask + finetune (bottom row) hits 1.0 on every task and beats single-task everywhere; single-task rows degrade sharply off-diagonal.

Headline Numbers

2.73

Maze2D reward · fine-tuned (CL 5)
vs 1.13 for Decision Transformer

- **2.73** RC reward (vs 1.49 DT)
- **1** model, all 8 tasks
- **5** seeds, 1000 rollouts